

Three vertex models for epithelial morphogenesis in Plexus2

SPV, mid-surface AVM, and the monolayer AVM behind `mono_V4_tube_flat`

Plexus2 prototype note — `prototype/Tyssue`

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Abstract

This note fixes, in one place, the mathematics of the three cell-mechanics models the Plexus2 tissue prototype uses, so that the `mono_V4_tube_flat` simulation — a flat epithelium that grows a tube by localized proliferation — can be studied on a sound footing. We give the energy, the equations of motion, the topological update, and the Plexus2 operator for each: (A) the *Self-Propelled Voronoi* (SPV) model (cells are points), (B) the *mid-surface Active Vertex Model* (AVM) used for the vesicle tubing runs (cells are surface polygons with a lumen-wedge volume), and (C) the *monolayer AVM* (cells are prisms with their own apical/basal surfaces and a true 3D volume), which `mono_V4` is built on. We then state precisely how the three differ, and set up the study of `mono_V4`.

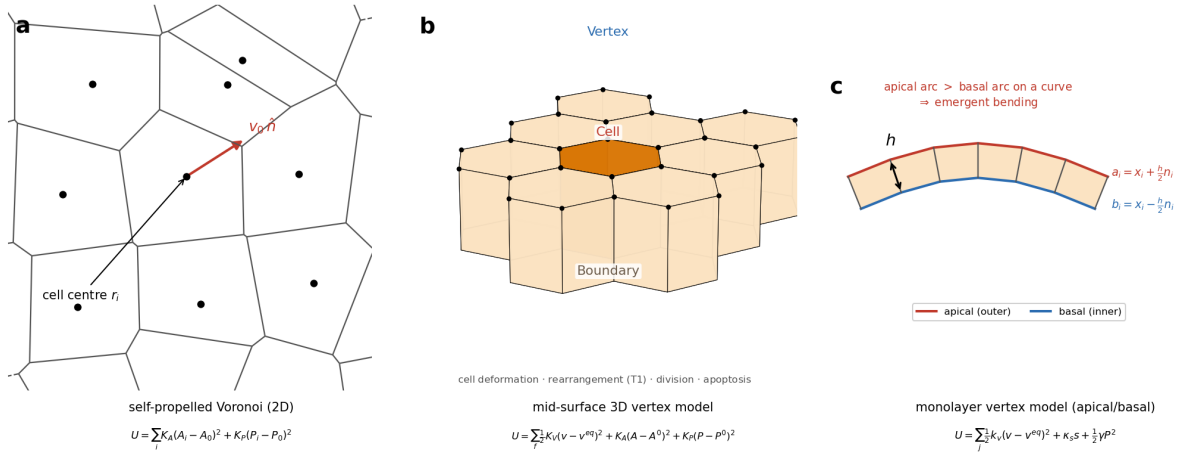


Figure 1: The three cell-mechanics models. (A) Self-Propelled Voronoi: a cell is a *point* r_i ; its polygon is the derived Voronoi cell; T1 is implicit (retessellate); motion adds self-propulsion $v_0 \hat{n}$. (B) Mid-surface AVM: the mesh *vertices* x_i are the DOF, cells are faces sharing junctions; the per-cell volume is a lumen *wedge* $v_f = \frac{1}{3} c_f \cdot N_f$ from the shell centre O ; T1 is an explicit flip. (C) Monolayer AVM (cross-section): each cell is a *prism* with an apical (outer) and basal (inner) surface offset $\pm h/2$ along the vertex normal, a true volume $A^{mid} h$ and surface $A^{ap} + A^{ba} + A^{lat}$; on a curved sheet the apical arc exceeds the basal arc, so surface tension penalises curvature — bending is *emergent*.

1 Common ground

All three are *overdamped* (inertia-free) cell-mechanics models: a mechanical energy U assigns a force to each degree of freedom, and the tissue moves down that force. They differ in (i) *what the degrees of freedom are*, (ii) *how a cell’s shape/volume is measured*, and (iii) *how neighbour exchange (T1) happens*. The overdamped law is, generically,

$$\eta \frac{d\mathbf{q}}{dt} = - \frac{\partial U}{\partial \mathbf{q}} + \mathbf{f}^{\text{active}}, \quad (1)$$

with $\mathbf{q}$ the DOF, η a friction, and $\mathbf{f}^{\text{active}}$ any non-conservative drive (self-propulsion, extrusion). Plexus2 integrates this with a *bounded* explicit Euler step (each displacement capped at a fraction of the mean edge length) so the rollout stays stable *and* differentiable.

What is a degree of freedom. A *degree of freedom* (DOF) is one independent coordinate the solver is allowed to move; the state vector $\mathbf{q}$ simply stacks all of them, and *every* geometric quantity a model computes — a cell’s area, volume, surface, the energy U , and hence the force $-\partial U/\partial \mathbf{q}$ — is a function of $\mathbf{q}$ alone. Choosing the DOF is therefore the first modelling decision, and it is exactly where the three models part ways: model A takes the DOF to be the cell-centre *points* $\mathbf{r}_i$ (two numbers per cell), while models B and C take them to be the mesh *vertices* $\mathbf{x}_i$ (three numbers per shared junction). Everything else — how a cell’s shape is measured, how neighbours are exchanged — follows from that choice.

Neighbour exchange (T1)

A tissue flows by cells swapping neighbours, not by cells changing volume. The elementary move is the *T1 transition* (Fig. 2): a short cell–cell interface shrinks to a point (a transient four-fold vertex) and re-opens rotated by 90° . The two cells that shared the edge lose contact, and the two cells that were kept apart become neighbours. This is the only way an epithelium of fixed-volume cells can rearrange (fluidise) and the reason a high-aspect structure like a tube can form at all — the wall cells must slide past one another as it extends. The three models realise it very differently: in the SPV it is *implicit* (re-deriving the Voronoi tessellation each step performs the swap automatically the instant two centres’ cells stop touching); in the vertex models it is an *explicit* operation — an edge whose length falls below a threshold $\Delta \ell_{\text{th}}$ is surgically flipped (`reconnect_t1.3d`), keeping the half-edge mesh closed.

2 Model A — Self-Propelled Voronoi (SPV)

Degrees of freedom. Each cell i is a single *point* $\mathbf{r}_i$. The tissue is the *Voronoi tessellation* of the points; cell i ’s polygon (area A_i , perimeter P_i) is a derived quantity, recomputed every step.

Energy. The standard 2D vertex-model shape energy evaluated on the Voronoi cells:

$$U_{\text{SPV}} = \sum_i \left[K_A (A_i - A_0)^2 + K_P (P_i - P_0)^2 \right], \quad p_0 \equiv \frac{P_0}{\sqrt{A_0}} \text{ (target shape index)}. \quad (2)$$

The dimensionless target shape index p_0 sets a rigidity (jamming) transition: the tissue is a *solid* (finite energy barrier to rearrangement) for $p_0 \lesssim 3.81$ and a *fluid* above it [1, 2].

Dynamics. Overdamped motion plus a persistent self-propulsion of fixed speed v_0 along a heading θ_i that rotationally diffuses (coefficient D_r):

$$\frac{d\mathbf{r}_i}{dt} = \mu \left(- \nabla_i U_{\text{SPV}} \right) + v_0 \hat{\mathbf{n}}(\theta_i), \quad \frac{d\theta_i}{dt} = \sqrt{2D_r} \xi_i(t). \quad (3)$$

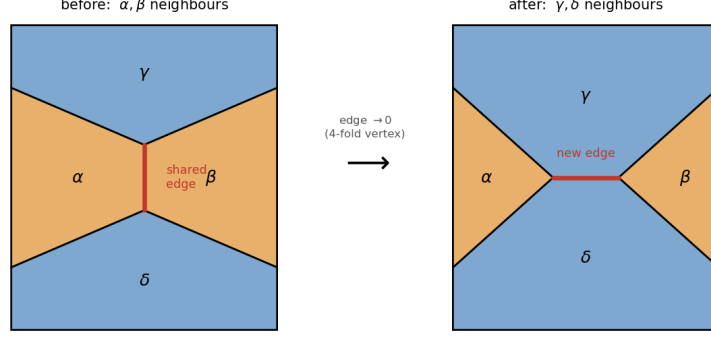


Figure 2: T1 transition (neighbour exchange). *Before*: cells α, β (orange) share the short red edge; γ, δ (blue) are separated by it. The edge contracts to a four-fold vertex and reopens perpendicular. *After*: γ, δ now share the edge and α, β are pushed apart. No cell changes area; only the connectivity changes.

T1 is implicit. Because the polygons are re-derived from the points each step, a neighbour exchange happens automatically the instant two points' Voronoi cells stop sharing an edge — there is no explicit surgery. This is the model's great convenience and its central limitation: the topology is slaved to the point positions, so it cannot represent a configuration the Voronoi rule forbids (e.g. a high-aspect junctional tube, a lumen, a fourfold vertex held open).

Plexus2 implementation. `prototype/Turing.vertex: operator voronoi_tension (2D)/vertex_tension` on a cell set of points; the Voronoi is rebuilt each tick; parameters `p0`, `K_A`, `K_P`, `mu`, `v0`, `Dr`. Reaction–diffusion couples on the Delaunay (neighbour) graph.

3 Model B — mid-surface Active Vertex Model (the tubing runs)

Degrees of freedom. The *vertices* $\mathbf{x}_i \in \mathbb{R}^3$ of an explicit half-edge mesh. A cell is a *face* (an ordered ring of vertices); neighbouring faces *share* vertices and edges. This is the true vertex model (AVM): the junctional network itself is the state, not a derived tessellation.

Geometry. Per face f : area vector by the Newell formula $\mathbf{N}_f = \frac{1}{2} \sum_{e \in f} \mathbf{x}_{s(e)} \times \mathbf{x}_{t(e)}$, area $A_f = \|\mathbf{N}_f\|$, perimeter $P_f = \sum_{e \in f} \ell_e$, and a *lumen-wedge volume* — the signed volume of the pyramid from the shell centre to the face,

$$v_f = \frac{1}{3} \mathbf{c}_f \cdot \mathbf{N}_f, \quad \mathbf{c}_f = \text{face centroid.} \quad (4)$$

The wedge volume is a *proxy* for the cell's contribution to the enclosed lumen; each cell carries its own, so the term below resists local buckling (a distributed constraint, not one global lumen scalar).

Energy.

$$U_{\text{AVM}} = \sum_f \left[K_A (A_f - A_f^0)^2 + K_P (P_f - P_f^0)^2 + \frac{1}{2} \Gamma P_f^2 + K_V (v_f - v_f^{\text{eq}})^2 \right] + \Lambda \sum_e \ell_e + K_R \sum_i (\|\mathbf{x}_i\| - R_0)^2. \quad (5)$$

Terms: area elasticity, perimeter elasticity, cortical contractility $\frac{1}{2} \Gamma P^2$, per-cell volume elasticity K_V , line tension Λ on edges, and an optional soft radial anchor K_R [3, 4, 5].

Dynamics & topology. Force $\mathbf{F}_i = -\partial U_{\text{AVM}} / \partial \mathbf{x}_i$ by one autograd pass; bounded overdamped Euler, iterated `relax_iters` times per frame toward force balance (Okuda relaxes to a residual

threshold E_{th} instead — see §4). T1 is *explicit*: `reconnect_t1_3d` flips any interior edge shorter than $\Delta\ell_{\text{th}}$ by a local four-cell neighbour exchange, keeping the mesh closed (Euler = 2). Division splits a face by an edge-midpoint septum *normal to the cell’s longest axis*; the reversible network reconnection (RNR) of [4] is the 3D generalization.

Plexus2 implementation. `shape_energy_3d` (the “default” implementation of the `shape_energy_3d` contract), `reconnect_t1_3d`, `divide_3d`, on a `vertex` set carrying the half-edge table. This is the model used for the vesicle tubing runs (`round.*`).

4 Model C — monolayer AVM (behind `mono_V4_tube_flat`)

Motivation. The mid-surface model represents a cell only by a polygon plus a wedge volume; it has no *thickness*, so it cannot express the apical/basal distinction, a true cell volume, or the bending rigidity of a monolayer. Okuda’s actual 3D model [5, 6] gives every cell its own volume and surface. Model C is the minimal lift of the AVM to that: keep the mid-surface vertices as the only DOF, but give each cell a *thickness* and read off an apical and a basal surface.

Geometry. For thickness h_j per cell and the area-weighted *vertex normal* $\mathbf{n}_i = \widehat{\sum_{f \ni i} \mathbf{N}_f}$, define the apical (outer) and basal (inner) vertices

$$\mathbf{a}_i = \mathbf{x}_i + \frac{H_i}{2} \mathbf{n}_i, \quad \mathbf{b}_i = \mathbf{x}_i - \frac{H_i}{2} \mathbf{n}_i, \quad H_i = \text{mean of } h_j \text{ over cells at } i. \quad (6)$$

Each cell is then a *prism* (apical cap + basal cap + lateral quads), with

$$v_j = A_j^{\text{mid}} h_j \quad (\text{cell volume}), \quad s_j = A_j^{\text{ap}} + A_j^{\text{ba}} + A_j^{\text{lat}} \quad (\text{cell surface}), \quad (7)$$

where $A_j^{\text{ap}}, A_j^{\text{ba}}$ are the Newell areas of the offset caps and A_j^{lat} sums the lateral quad areas. *Emergent bending*: because the caps are offset along the *vertex* normal, a curved sheet has $A^{\text{ap}} \neq A^{\text{ba}}$ (the convex side stretches), so surface tension κ_s penalises curvature with *no explicit bending term* — the apical/basal area ratio matches $((R+h/2)/(R-h/2))^2$ to 0.2% in the validation.

Energy (Okuda Eq. 3, plus a shape regularizer).

$$U_{\text{mono}} = \sum_j \left[\frac{1}{2} k_v (v_j - v_j^{\text{eq}})^2 + \kappa_s s_j + \frac{1}{2} \gamma P_j^2 \right] + \Lambda \sum_e \ell_e + K_R \sum_i (||\mathbf{x}_i|| - R_0)^2. \quad (8)$$

The first two terms are Okuda’s per-cell volume elasticity and *linear surface tension*. The $\frac{1}{2}\gamma P^2$ contractility is a *cell-shape regularizer* standing in for the continuous RNR/T1 remeshing Okuda relies on: without it the bare volume+surface energy shears or spikes under large deformation.

Per-cell stress (from the same energy). The isotropic pressure and cortical tension read off Eq. 8 are

$$p_j = -\frac{\partial U}{\partial v_j} = 2k_v (v_j^{\text{eq}} - v_j) \quad (> 0 : \text{below-target volume, pushes out}), \quad T_j = \frac{\partial U}{\partial P_j} = \gamma P_j + \Lambda. \quad (9)$$

These are the fields `analyze_forces.py` plots; the whole energy is torch-differentiable, so a loss on stress can be optimised.

`mono_V4_tube_flat` specifics. The substrate is a *flat* open epithelium (`build_flat_mesh`: a jittered planar Voronoi patch, boundary rim *pinned*). A central *growth zone* that rides the advancing tip has its target volume v^{eq} raised each round; the tissue relaxes under Eq. 8 (bounded Euler, tangential Lloyd smoothing); oversized cells *divide in-plane*, the septum normal to the longest axis (Hertwig). Localized in-plane proliferation on a rim-pinned sheet has nowhere to go but *buckle out*

of *plane*, extending a tube whose wall is the new cells. There is no reaction–diffusion here — the growth zone is prescribed geometrically.

Plexus2 implementation. `shape_energy_3d` with `implementation=monolayer` (class `MonolayerShapeEnergy3D`) — the plexus2 “same biology, different numerics” pattern: the monolayer is a second numerical implementation of the same shape-energy operator contract. Geometry: `monolayer_geometry_3d`; flat substrate: `flat_mesh.py`; driver: `demo_monolayer.py` (`demo_tube`). Division reuses the mid-surface topology ops (`divide_face_3d`).

5 How the three differ

	A. SPV	B. mid-surface AVM	C. monolayer AVM
DOF	cell centres $\mathbf{r}_i$	mesh vertices $\mathbf{x}_i$	mesh vertices $\mathbf{x}_i$ ($+h_j$)
cell shape	Voronoi polygon (derived)	face polygon (explicit)	prism: apical+basal+lateral
volume	— (2D)	lumen wedge $\frac{1}{3}\mathbf{c}\cdot\mathbf{N}$	true $A^{\text{mid}}h$
surface	perimeter only	area+perimeter	apical+basal+lateral
bending	—	none (or explicit K_{bend})	<i>emergent</i> (cap offset)
T1 / neighbour exch.	implicit (retessellate)	explicit RNR/T1 flip	explicit (inherits B’s)
energy	Eq. 2	Eq. 5	Eq. 8 (Okuda 3)
can hold a tube?	no (Voronoi-forbidden)	yes (junctional net)	yes, with a true wall

The essential distinctions. *A vs B*: SPV’s topology is a slave of the point positions (implicit T1), so it is fast and always valid but cannot represent a high-aspect tube or a lumen; the AVM makes the junctional network the state, paying for explicit T1 surgery but able to hold a tube. *B vs C*: the mid-surface AVM has no thickness — its “volume” is a wedge from the shell centre and it has no apical/basal surfaces or bending; the monolayer AVM gives each cell a real volume Ah and a real surface, so localized volume growth *buckles* into a tube as an *equilibrium* shape (Okuda’s growth-driven, quasi-static route), whereas the mid-surface tube in the `round_*` runs is held out by an added extrusion force (a forced, higher-stress protrusion).

6 Plexus2 implementation: sets, operators, fields

Plexus2 holds the tissue state as named *sets* of elements, each carrying per-element *fields*, and advances it with *operators* that read those fields and write forces or topology. The three models map onto that structure as below (operator names are the registered `op` identifiers in the prototype).

	Sets	Fields (per element)	Operators (op)
A. SPV	one cell set (points); neighbour graph <i>derived</i> (De-launay)	pos $\mathbf{r}_i$, heading θ_i ; rest $A0, P0$; morphogen c_i	<code>cell_geometry</code> (Voronoi A, P), <code>shape_energy</code> , <code>cell_adjacency</code> , <code>t1_transition</code> (implicit, re-tessellation), <code>face_divide</code> , <code>cell_react/diffuse</code>
B. mid-surface AVM	vertex set + half-edge table (es,et,ef) defining the face =cell set	vertex pos $\mathbf{x}_i$; per-face $A0, P0, v_{eq}, hc$; per-edge ℓ_0 ; per-cell morphogen	<code>cell_geometry_3d</code> , <code>shape_energy_3d</code> (default), <code>reconnect_t1_3d</code> , <code>divide_3d</code> , <code>morphogen_growth_3d</code> , <code>topo_snapshot_3d</code>
C. mono-layer AVM	same vertex + face sets; apical/basal are <i>derived</i> , never stored	vertex pos $\mathbf{x}_i$; per-cell thickness h , v_{eq} , morphogen <i>amount</i> m_j	<code>shape_energy_3d</code> with <code>implementation=monolayer</code> (MonolayerShapeEnergy3D), <code>reconnect_t1_3d</code> , <code>divide_3d</code> , <code>morphogen_growth_3d</code> (<code>conserve_amount</code>), <code>cell_rd_seed</code>

Three points are worth stressing. (i) *The set choice is the DOF choice*: A stores a **cell** set of points, so its only positional field is $\mathbf{r}_i$; B and C store a **vertex** set, so the positional field is $\mathbf{x}_i$ and cells are read off the shared half-edge table. (ii) *Same operator, two implementations*: B and C are both the `shape_energy_3d` operator — C is a second numerical implementation (`implementation=monolayer`) of the *same* contract, so the surrounding pipeline (growth, T1, division, rollout) is untouched. (iii) *The monolayer adds exactly one field*, the per-cell thickness h : the apical/basal vertices $\mathbf{a}_i, \mathbf{b}_i$ are recomputed from $(\mathbf{x}_i, \mathbf{n}_i, h)$ every step, never stored — which is what makes the lift from B to C cheap and keeps the state differentiable.

7 The mono_V4_tube_flat study (what we will make sound)

`mono_V4` currently extends a lumpy, chaotic tube: growth is confined to a moving tip cap, cells divide when oversized, and the sheet buckles — but the wall is irregular and the dynamics are not understood. The plan is to make this one simulation rigorous:

1. **Start from a working point.** `mono_V3b_buckle_flat` (a clean localized buckle, modest amplitude) is a controlled, understood equilibrium; growing the tube *from* that buckle — rather than the current fast, uncontrolled extension — is the sound way in.
2. **Levers.** κ_s/k_v (surface tension vs volume stiffness $\rightarrow$ buckle amplitude and wall tension), h_0 (thickness $\rightarrow$ emergent bending, tube diameter), γ (cell rounding), the growth rate and division threshold (quasi-static vs explosive), and the tip-cap size (tube diameter).
3. **Analysis.** Run `analyze_forces.py` throughout: per-cell force $\|-\nabla U\|$, pressure p_j , cortical tension T_j , cell migration and division — to read the wall stress and check the growth-driven regime is quasi-static (bounded, tube $\leftrightarrow$ body-shared stress) rather than explosive.

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